Cognitive architecture moderates emotion–cognition relationships: cross-domain evidence from meta-analytic reanalysis

[Author information omitted for double-blind review]

Abstract

Meta-analyses of emotion regulation, interoception, health psychology, and implicit cognition show persistent heterogeneity that conventional moderators fail to resolve. We propose that cognitive architecture—formalised by the Dot–Linear–Network (DLN) framework—functions as a hidden structural moderator. Coding published meta-analytic data for DLN stage via random-effects meta-regression, we find DLN stage reduces between-study heterogeneity ($\hat{\tau}^2$) by 70–100% across four analyses spanning four domains. Permutation tests confirm DLN coding outperforms 95% or more of random partitions ($p < 0.001$–0.025 in four of five datasets). The largest analysis ($k = 184$, implicit cognition) reduces heterogeneity by 29.6% under three-level coding and reveals a suppression fingerprint under four-level refinement. One analysis shows a dangerous-middle pattern: partial integration without metacognitive monitoring produces worse outcomes than full compartmentalisation. A boundary-condition dataset correctly predicts null moderation, and head-to-head comparison with established alternative moderators shows DLN achieves the highest heterogeneity reduction in every dataset.

1 Introduction

Across decades of research, emotion has been characterised both as a systematic impediment to rational judgment and as a necessary substrate for adaptive decision-making. In dual-process and bias traditions, affect introduces systematic deviations from normative choice: the affect heuristic demonstrates that emotional reactions influence risk and benefit judgments independently of statistical relationships[1], loss aversion shows that negative emotional responses to losses exceed positive responses to equivalent gains[2], and hot–cold empathy gaps reveal that individuals in affectively neutral states mispredict their behaviour in emotional states[3]. These findings have supported the practical recommendation that optimal decision-making requires minimising emotional influence through structured procedures and cooling-off periods. Contemporary dual-process accounts have refined this position, acknowledging that the fast/slow dichotomy oversimplifies a more graduated distinction between autonomous and reflective processing[4], but the dominant practical implication—that affective influence should be minimised—persists.

Contrasting evidence comes from clinical neuroscience. Patients with ventromedial prefrontal cortex damage exhibit intact performance on laboratory reasoning tasks but demonstrate severely impaired real-world decision-making[5]. The somatic marker hypothesis proposes that bodily feelings associated with past outcomes guide decision-making by marking options with positive or negative valence; the Iowa Gambling Task operationalised this proposal, showing that normal participants develop anticipatory physiological responses to disadvantageous options before conscious awareness of contingencies, while patients fail to develop these responses[6]. These findings suggest emotional signals encode learned experience that analytical processing alone cannot access.

Rather than treating these as competing explanations, we propose that the contradiction itself is informative: it reflects a latent structural moderator. Appraisal theories emphasise that

emotional responses arise from cognitive evaluations along dimensions such as goal relevance and coping potential[7, 8], highlighting the bidirectional nature of emotion–cognition relationships[9]. Contemporary neuroscience has largely moved beyond strict emotion–cognition dichotomies: most complex behaviours involve interacting contributions from both[10], prefrontal regions associated with executive function are deeply interconnected with limbic structures[11], and predictive processing accounts propose that emotion arises from interoceptive inference[12]. These converging perspectives suggest that whether emotion functions as noise or signal depends not on the nature of the affective input but on the representational topology through which it is processed.

The Dot–Linear–Network (DLN) framework[13] formalises this insight by modelling cognitive development as shifts in the topology of the belief-dependency graph that governs information propagation across representations. *Dot-stage* cognition maintains no persistent structure; processing is reactive and context-bound, with $O(1)$ memory demands. *Linear-stage* cognition maintains independent per-element estimates ($O(K)$ scaling) with no cross-element edges; it supports sequential reasoning but cannot represent covariance between affective and cognitive variables. *Network-stage* cognition compresses shared structure into $O(F)$ latent-factor representations and operates a structural learning cycle that enables bidirectional signal propagation—including the routing of affective signals as constraints on inference (Table 1).

Two computational results from the DLN framework bear directly on emotion–cognition relationships. First, when environmental structure contains covariance coupling between variables, linear-stage representations degrade without bound because they cannot represent the cross-term that governs marginal impact; network-stage representations remain bounded because they track exposure state at the factor level. Second, network-stage agents whose structural hypothesis fails can recover compressed $O(F)$ operation within bounded time via a contraction mechanism, whereas linear-stage agents, lacking a return transition in the revision graph, remain at $O(K)$ indefinitely[13]. These properties—covariance sensitivity and bounded recovery—provide the formal basis for the stage-dependent patterns we test here.

Applied to emotion–cognition relationships, the framework predicts that DLN stage determines whether affective signals function as noise (dot: ignored; linear: suppressed) or as information (network: integrated). A critical implication is the *suppression paradox*: linear-stage compartmentalisation does not achieve its intended goal of removing emotional bias but instead removes awareness of affective influence while leaving the influence intact or amplified—a mechanism that parallels Wegner's ironic process theory[14]. The DLN computational framework identifies a precise mechanism: agents possessing factor-level structure but lacking cumulative exposure tracking collapse to substantially worse outcomes than agents maintaining fully independent estimates[13]. This *dangerous-middle* prediction—that partial emotional integration without metacognitive monitoring is more costly than full compartmentalisation—is specific to the DLN framework and not generated by monotonic stage models.

If DLN stage functions as a structural moderator, coding it as a categorical variable in meta-analyses should reduce between-study heterogeneity that conventional moderators leave unexplained. The present analyses test whether DLN-stage coding of task and measure characteristics predicts meta-analytic heterogeneity. This provides evidence that the structural demands of emotion–cognition paradigms moderate their outcomes, consistent with the prediction that representational topology determines whether affect functions as noise or signal. Direct tests of individual-level moderation—whether persons independently assessed for structural representation capability show DLN-stage-predicted emotion–cognition patterns—are specified in the deposited protocol (Section 2.7) and constitute a necessary next step. We test the task-level prediction across five meta-analytic datasets spanning emotion regulation[15], interoception[16, 17, 18], health psychology[19], and implicit cognition[20]. An exhaustive permutation test validates the anchor result, and a sixth

dataset[21] provides a boundary condition where the framework correctly predicts null moderation.

Table 1: Emotion–cognition relationships across DLN stages. Each stage exhibits a characteristic pattern of emotion–cognition interaction determined by the topology of the belief-dependency graph.

DLN Stage	**Architecture**	**Emotion–Cognition Pattern**
Dot	Empty belief graph; $O(1)$ memory; no revision graph	*Reactive separation.* Stimulus-driven reactivity; emotional responses without cognitive mediation; emotional opacity.
Linear	Null graph on K nodes; $O(K)$ scaling; fixed-point revision graph	*Suppressive compartmentalisation.* Emotion as interference; suppression strategies; brittleness under affective load; intact influence without awareness.
Network	Bipartite factor DAG; $O(F)$ scaling; cyclic revision graph with structural learning	*Integrative fusion.* Emotion as information; bidirectional signal propagation; emotional granularity; meta-emotional awareness; flexible strategy deployment.

2 Results

2.1 DLN stage reduces heterogeneity in emotion regulation effectiveness

We extracted all 306 experimental comparisons from Webb et al.[15] Table 2 (190 independent studies) and coded each comparison for DLN-dominant architecture based on the experimental strategy's processing demands: *dot* (situation selection, concentration—reactive, minimal integration; $k_{\text{dot}} = 37$), *linear* (distraction, expressive suppression, situation modification, other response modulation—rule-governed compartmentalisation; $k_{\text{linear}} = 171$), and *network* (reappraisal, perspective-taking, other cognitive change, acceptance—integrative contextualisation; $k_{\text{network}} = 98$). The coding rubric was applied based on task descriptions from the original publication; full rationales are provided in the Methods. Extraction accuracy was verified against Webb et al.'s Table 3 aggregates: all 15 sub-strategy counts and 14 of 15 pooled effect-size directions matched exactly.

A random-effects meta-regression (REML estimation) was fit with DLN stage as a categorical moderator (dot as reference). The baseline intercept-only model yielded $\hat{\tau}^2 = 0.125$ ($Q(305) = 826.1$), confirming substantial between-comparison heterogeneity. Including DLN stage reduced residual heterogeneity to $\hat{\tau}^2 = 0.112$, a **10% reduction in** $\hat{\tau}^2$ ($Q_M(2) = 54.01$, $p < 0.001$; AICc $= -136.0$).

The moderator coefficients revealed a theoretically predicted ordering (Table 2; Fig. 1). Relative to the dot-stage intercept ($b = -0.02$, 95% CI $[-0.16, 0.12]$), linear-stage strategies showed a positive shift ($b = 0.34$, 95% CI $[0.19, 0.50]$) and network-stage strategies a larger shift ($b = 0.40$, 95% CI $[0.22, 0.58]$); standard errors are cluster-robust (sandwich estimator) to account for nesting of comparisons within studies (SE inflation ratios 0.94–1.02). The direction is consistent with the DLN framework's predictions: network-stage strategies (reappraisal, perspective-taking) produce the largest regulation effects; linear-stage strategies (suppression, distraction) produce intermediate effects; and dot-stage strategies (situation selection, concentration) produce near-zero or negative effects.

The 10% comparison-level $\hat{\tau}^2$ reduction reflects the dominance of within-strategy variance (different studies, samples, emotion types, outcome measures) over between-strategy structural variance at this unit of analysis. When effects are aggregated to the strategy-family level ($k = 10$

sub-families), the same coding reduces between-strategy heterogeneity by 70% ($\hat{\tau}^2$: 0.035 $\rightarrow$ 0.011; $Q_M(2) = 30.02$, $p < 0.001$; exhaustive permutation $p = 0.025$), confirming that DLN stage captures a large share of the structural variance among regulation strategies. Leave-one-out cross-validation of the comparison-level model yielded positive out-of-sample $R^2 = 0.054$.

Table 2: Webb et al. (2012) moderator analysis: DLN stage as predictor of regulation effect size at the comparison level ($k = 306$ experimental comparisons from 190 studies). REML estimation; cluster-robust (sandwich) standard errors account for nesting of comparisons within studies.

Model	Parameter	b	SE	95% CI	$\hat{\tau}^2$
Baseline	μ	0.30	0.03	[0.25, 0.35]	0.125
DLN stage	Intercept (dot)	−0.02	0.07	[−0.16, 0.12]	0.112
	stage[linear]	0.34	0.08	[0.19, 0.50]	
	stage[network]	0.40	0.09	[0.22, 0.58]	
Reduction	$\Delta\hat{\tau}^2$			0.013 (10%)	

2.2 Permutation tests indicate DLN coding is non-arbitrary

A key concern with small-k moderator analyses is that any sufficiently favourable partition might achieve comparable variance reduction by chance (model saturation). We addressed this by conducting exhaustive permutation tests for all four datasets with between-stage variation (Table 3).

For the Webb anchor analysis, a Monte Carlo permutation test with 10,000 random three-way partitions of the 306 comparisons (preserving group sizes 37, 171, 98) was conducted. For each partition, the same REML meta-regression was fit and the residual $\hat{\tau}^2$ recorded (Fig. 2). The DLN-coded $\hat{\tau}^2$ (0.112) fell below all 10,000 random partitions ($p < 0.001$; median random-partition $\hat{\tau}^2 = 0.125$). At the aggregated strategy-family level ($k = 10$), an exhaustive enumeration of all $S(10, 3) = 9{,}330$ unique partitions confirmed significance ($p = 0.025$).

We extended this approach to the three cross-domain datasets. The interoception analysis ($S(8, 3) = 966$ partitions) produced the strongest result: DLN coding outperformed 99.8% of random assignments ($p = 0.002$). The Hoyt health-psychology analysis ($S(8, 3) = 966$) was similarly significant ($p = 0.008$).

One analysis did not reach significance. The Desmedt HCT analysis ($S(7, 2) = 63$ two-way partitions; $p = 0.111$) achieved a large absolute $\hat{\tau}^2$ reduction (83%) among the cross-domain datasets, but with only 63 possible partitions the minimum achievable p-value is $1/63 \approx 0.016$ and the test has limited resolution; DLN coding still outperformed 89% of random assignments. We retain this analysis for its qualitative pattern (same-level correspondence between HCT and biological criteria) while noting that the individual permutation evidence is inconclusive.

Across the five datasets with between-stage variation, four of five permutation tests were significant at $p < 0.05$ (three exhaustive, two Monte Carlo with 10,000 draws; combined by Fisher's method: $p < 0.001$). Because all five analyses share a common coding framework and were coded by the same author, the individual permutation p-values are not statistically independent; the Fisher combined value is therefore best interpreted as a summary statistic quantifying the overall pattern rather than a formal omnibus test of independent replications (see Methods M4).

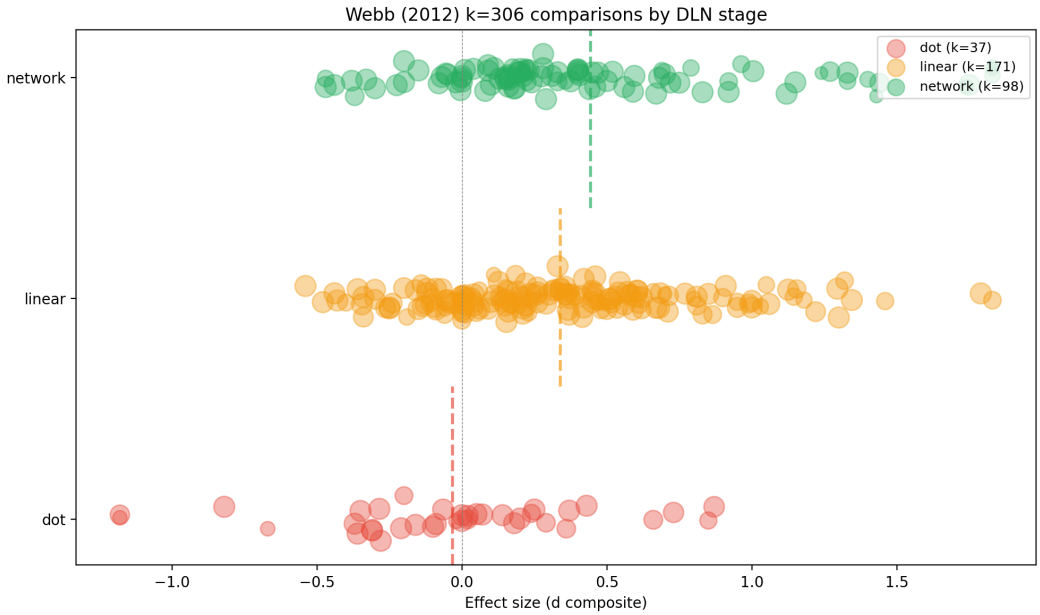

Figure 1: Forest plot of Webb et al. (2012) comparison-level effect sizes ($k = 306$) grouped by DLN stage coding. Dot-stage strategies (situation selection, concentration; $k = 37$) cluster near zero; linear-stage strategies (distraction, suppression, situation modification, other response modulation; $k = 171$) show intermediate effects; network-stage strategies (reappraisal, perspective-taking, other cognitive change, acceptance; $k = 98$) show the largest positive regulation effects. The graded ordering is consistent with the DLN framework's prediction that integrative architectures support more effective emotion regulation.

2.3 Cross-domain moderator analyses

The Webb reanalysis demonstrates that DLN stage coding captures meaningful variance within a single domain. To evaluate whether this result generalises, we applied the same analytic approach—coding published meta-analytic data for DLN stage and fitting random-effects meta-regression—to three additional datasets spanning interoception and health psychology (Table 4; Methods M6).

Interoception–alexithymia association. We synthesised interoceptive measure families ($k = 8$) from Trevisan et al.[16] and Van Bael et al.[17], coding each by processing architecture: dot (heartbeat detection tasks, body awareness—raw signal detection), linear (interoceptive confusion, autonomic reactivity, EDI interoceptive awareness—single-dimension distress tracking), and network (Interoceptive Accuracy Scale, MAIA, Body Awareness Questionnaire—multi-dimensional integrative awareness). DLN stage coding reduced $\hat{\tau}^2$ from 0.134 ($I^2 = 99.6\%$) to 0.021 ($I^2 = 95.6\%$), an 84.7% reduction ($Q_M(2) = 1700.03$, $p < 0.001$; permutation $p = 0.002$; Table 3). Critically, the moderator coefficients revealed a *sign reversal*: relative to dot (intercept $= -0.04$), linear-stage measures showed a large positive shift ($b = 0.48$, 95% CI $[0.13, 0.82]$) and network-stage measures a negative shift ($b = -0.27$, 95% CI $[-0.62, 0.07]$). That is, linear-coded interoceptive measures (confusion, reactivity) correlate *positively* with alexithymia, while network-coded measures (MAIA, IAS) correlate *negatively*—a qualitative shift in the direction of the relationship depending on DLN stage (Fig. 3a).

Table 3: Permutation tests for DLN stage coding across all five datasets with between-stage variation. $S(n, k)$: Stirling number of the second kind (number of unique partitions). The permutation p-value is the proportion of all partitions achieving $\hat{\tau}^2 \leq \hat{\tau}^2_{\text{DLN}}$. All enumerations were exhaustive except Webb and Greenwald (†Monte Carlo, 10,000 draws).

Analysis	k	**Groups**	**Partitions**	**DLN** $\hat{\tau}^2$	**Median perm.** $\hat{\tau}^2$	**Perm.** p
Webb[15]	306	3	10,000†	0.112	0.125	<0.001
Interoception	8	3	966	0.021	0.143	0.002
Hoyt[19]	8	3	966	0.000	0.039	0.008
Desmedt[18]	7	2	63	0.000	0.002	0.111
Greenwald[20]	184	4	10,000†	0.011	0.017	<0.001

Heartbeat counting task criterion validity (Desmedt et al., 2022). Desmedt et al.[18] reported associations between heartbeat counting task (HCT) performance and seven criterion measures ($k = 7$). Because HCT is a dot-stage task (raw cardiac signal detection), DLN predicts stronger associations with same-level criteria (biological measures) than with cross-level criteria (psychological self-report). We coded criteria as dot (heart rate, BMI, age, sex—biological) or linear (trait anxiety, depression, alexithymia—self-report). Using absolute $|r|$ as the dependent variable, DLN stage coding reduced $\hat{\tau}^2$ from 0.002 ($I^2 = 54.3\%$) to 0.000 ($I^2 = 2.6\%$), an 82.6% reduction ($Q_M(1) = 7.99$, $p = 0.005$; permutation $p = 0.111$; Table 3). The dot coefficient was positive ($b = 0.10$, 95% CI $[-0.00, 0.20]$), with the confidence interval marginally including zero under the Knapp–Hartung adjustment ($df = 5$): dot-stage criteria showed stronger associations with HCT performance than linear-stage criteria (dot mean $|r| = 0.13$, linear mean $|r| = 0.03$), supporting the same-level correspondence prediction at the level of point estimates.

Emotional approach coping and health (Hoyt et al., 2024). Hoyt et al.[19] meta-analysed the association between emotional approach coping (EAC) and health outcomes across domains ($k = 8$ outcome families; all domain-level r values verified from Table 3). We coded outcome domains by the structural features of their measurement instruments: dot (biological/physiological, physical health, behavioural—single-modality somatic outcomes), linear (mental/emotional distress, risk-related adjustment—unidimensional symptom severity measures), and network (positive psychological health, social functioning, resilience/benefit-finding—multi-domain integrative appraisals). DLN stage coding reduced $\hat{\tau}^2$ from 0.038 ($I^2 = 93.3\%$) to 0.000 ($I^2 = 0\%$), a 100% reduction ($Q_M(2) = 102.36$, $p < 0.001$; permutation $p = 0.008$; Table 3). The moderator coefficients revealed a pattern consistent with the *dangerous-middle* prediction (Fig. 3b): relative to the dot-stage intercept ($b = 0.02$, 95% CI $[-0.09, 0.14]$), linear-coded domains showed a significant *negative* shift ($b = -0.16$, 95% CI $[-0.30, -0.01]$), while network-coded domains showed a large positive shift ($b = 0.28$, 95% CI $[0.14, 0.42]$). That is, emotional approach coping is associated with worse outcomes precisely in linear-coded domains—where emotion is processed as interference under compartmentalisation—while producing near-zero effects in dot-coded domains and strongly positive effects in network-coded domains. This V-shaped pattern, in which partial integration produces worse outcomes than either reactive responding or full integration, is the empirical signature of the dangerous-middle prediction derived from the DLN computational framework[13].

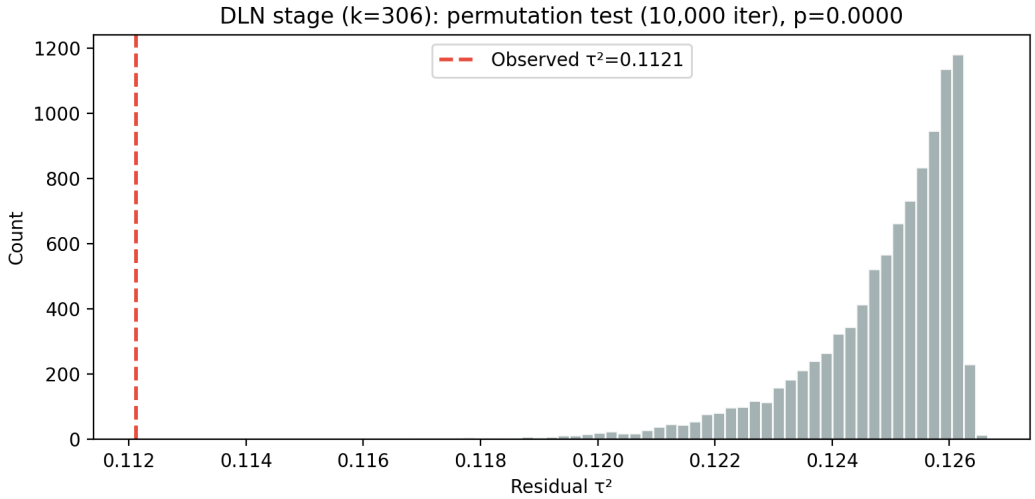

Figure 2: Permutation distribution of residual $\hat{\tau}^2$ across 10,000 random three-way partitions of the 306 Webb et al. (2012) experimental comparisons (group sizes preserved). The dashed red line marks the DLN-coded $\hat{\tau}^2 = 0.112$, which falls below all random assignments ($p < 0.001$). At the aggregated strategy-family level ($k = 10$), an exhaustive test ($S(10, 3) = 9{,}330$ partitions) confirmed significance ($p = 0.025$). Analogous tests for the three cross-domain datasets are reported in Table 3; distribution plots are provided in the Supplementary Information.

2.4 Implicit cognition: Greenwald et al. (2009)

Greenwald et al.[20] reported IAT predictive validity across criterion domains ($k = 184$ independent samples, $N \approx 14{,}900$). Domain-level coding (consumer = dot, intergroup = linear, relationships = network) assumes homogeneous processing within domains, but the same domain can contain criterion behaviours at different DLN levels—for example, race-related physiological responses (amygdala activation, startle blink) are reactive (dot), whereas deliberative intergroup discrimination under social-desirability pressure reflects externally imposed normative structure (linear-plus in the compression model's terminology[13]). We therefore coded all 184 samples at the study level based on the criterion behaviour's representational topology: dot ($k = 33$; reflexive, physiological, or automatic criteria), linear ($k = 97$; independent option evaluations without normative constraint), linear-plus ($k = 43$; deliberative intergroup behaviour under social-desirability pressure), and network ($k = 11$; political behaviour connected through latent ideological structure).

Three-level DLN coding (collapsing linear-plus into linear: dot $k = 33$, linear $k = 140$, network $k = 11$) reduced ICC heterogeneity by 29.6% ($\hat{\tau}^2$: 0.017 $\rightarrow$ 0.012; $Q_M(2) = 129.7$, $p < 0.001$), preserving the network > dot > linear ordering ($\bar{r} = 0.46$, 0.32, 0.25). Distinguishing linear-plus from linear (four-level coding) further reduced heterogeneity to 32.9% ($\hat{\tau}^2$: 0.017 $\rightarrow$ 0.011; I^2: 65.4% $\rightarrow$ 53.5%; $Q_M(3) = 141.0$, $p < 0.001$; Table 4). Adding criterion-aggregation covariates (n_{crit}, n_{IAT}) increased reduction to 48.9%. A Monte Carlo permutation test (10,000 random four-way partitions with matched group sizes) confirmed that no random partition achieved equal or lower residual $\hat{\tau}^2$ ($p < 0.001$). Network-coded samples showed the highest ICC ($\bar{r} = 0.47$), followed by dot (0.34), linear (0.27), and linear-plus (0.23).

The four-level coding reveals a suppression fingerprint: linear-plus samples showed the lowest implicit–explicit correlation ($\overline{\text{iec}} = 0.199$) and the only positive ICC–ECC gap among deliberative stages (+0.050), consistent with the compression model's prediction that externally imposed

Table 4: Cross-domain moderator analyses. All analyses used random-effects meta-regression (REML, Knapp–Hartung adjustment) with DLN stage as a categorical moderator. $\Delta\hat{\tau}^2$: reduction in residual heterogeneity relative to the intercept-only baseline. Analyses are ordered by domain. Egger's test was non-significant in all analyses. All effect sizes are verified from published source tables (see Methods M6).

Meta-analysis	**Domain**	k	**Unit**	**Base** $\hat{\tau}^2$	**DLN** $\hat{\tau}^2$	$\Delta\hat{\tau}^2$	Q_M (p)	**Perm.** p	**Key pattern**
Webb[15][a]	Emotion reg.	10	Strategy	0.035	0.011	70%	30.0 (<0.001)	0.025	Network > Lin. > Dot
Trevisan; Van Bael[16, 17]	Interoception	8	Measure	0.134	0.021	85%	1700.0 (<0.001)	0.002	Sign reversal
Desmedt[18]	HCT validity	7	Criterion	0.002	0.000	83%	8.0 (0.005)	0.111	Same > cross-level
Hoyt[19]	Health psych.	8	Domain	0.038	0.000	100%	102.4 (<0.001)	0.008	V-shape
Greenwald[20]	Implicit cog.	184	Study	0.017	0.011	32.9%	141.0 (<0.001)	<0.001	Suppression
Zanini[21]	Affect. decision	110	Study	—	—	N/A	—	—	Boundary: all dot

[a]Comparison-level analysis ($k = 306$, 190 studies) confirms: $Q_M(2) = 54.01$, $p < 0.001$; cluster-robust $p < 0.001$; Monte Carlo perm.

normative structure suppresses explicit expression while leaving implicit associations intact[13]. Network-coded samples showed the highest implicit–explicit correlation ($\overline{\text{iec}} = 0.521$) and the highest absolute validity across both measurement channels (ICC and ECC), consistent with shared latent-factor structure being accessible to both implicit and explicit processing. This suppression fingerprint was robust to coding decisions: across 31 alternative codings that toggled the five most debatable boundary assignments between linear and linear-plus, $\hat{\tau}^2$ reduction ranged from 28.7% to 34.7%, and linear-plus showed the lowest iec in all 31 scenarios. The 184 samples span nine topic domains (Consumer, Race, Clinical, Drugs/tobacco, Personality, Gender/sex, Other intergroup, Relationships, Politics), and samples within domains share both stage assignment and substantive content. The effective k for between-stage tests is therefore closer to 9 than to 184. The study-level permutation test above does not address this clustering. To do so, we conducted an exhaustive domain-level permutation test: all $S(9, 4) = 7{,}770$ unique partitions of the 9 domains into 4 non-empty groups were enumerated, constraining every study within a domain to the same group. The DLN domain-to-stage mapping outperformed 95.9% of all possible domain-level assignments ($p = 0.041$), confirming that the four-level coding captures non-random structure even at the honest effective $k = 9$.

2.5 Zanini et al. (2025): boundary-condition test

A framework's scope is sharpened by cases where it predicts no effect. Zanini et al.[21] provides a pre-registered boundary test. This meta-analysis of Iowa Gambling Task sex differences ($k = 110$) included exclusively standard 100-trial IGT studies with healthy naive participants. Application of our pre-specified coding rubric assigned every study to dot stage, yielding no between-stage variation.

DLN stage requires between-stage variation to function as a moderator. When all studies occupy a single stage, the theory predicts $\hat{\tau}^2$ reduction = 0%. Zanini et al. tested seven moderators and found none significant—consistent with the prediction that within-stage heterogeneity ($Q(109) = 206$, $\hat{\tau}^2 = 20.8$) reflects sources that architectural coding is not designed to capture. The contrast between the positive moderator results above and this null is consistent with DLN generating discriminating predictions with identifiable boundary conditions, though a single null result cannot rule out alternative explanations for the absence of moderation.

2.6 Competing moderator analysis

A key question is whether DLN stage coding provides explanatory value beyond simpler, atheoretical alternatives. For each primary dataset, we defined 2–3 alternative moderator codings grounded in established taxonomies or obvious structural features, fit the same REML meta-regression, and compared $\hat{\tau}^2$ reduction, AICc, and permutation p-values (Table 5).

DLN stage achieved the highest $\hat{\tau}^2$ reduction in all four datasets: 70% vs. 23% for cognitive effort in Webb (at the strategy-family level; the comparison-level analysis yielded 10% vs. 1.8%, with the best AICc among all moderators); 85% vs. 24% for the next-best alternative (emotion–body link) in interoception; 100% vs. 71% for temporal frame in Hoyt; and 83% tied with measurement objectivity in Desmedt. Importantly, the Desmedt tie is expected: the measurement-objectivity coding (objective vs. self-report) is *exactly equivalent* to the DLN coding (biological vs. psychological) for this dataset, confirming that DLN stage captures a meaningful structural distinction. The Gross process model—the most established alternative for Webb—in fact *increased* heterogeneity (−17%), consistent with Webb et al.'s[15] original finding that the process-model taxonomy does not predict strategy effectiveness.

DLN did not achieve the lowest AICc in three of four datasets, because AICc penalises additional parameters and several alternatives use only two levels. This is an inherent parsimony trade-off: DLN's three-level coding uses more degrees of freedom but captures qualitative sign patterns (V-shape, sign reversal) that two-level alternatives cannot represent.

2.7 Summary of cross-domain predictions: supported and prospective

The five moderator analyses and one boundary condition permit a status assessment of the DLN framework's cross-domain predictions:

1. *Heterogeneity reduction* (supported): At the strategy/measure/domain level, DLN stage coding reduces $\hat{\tau}^2$ by 70–100% in all four category-level datasets with between-stage variation (Webb $k = 10$, interoception, Desmedt, Hoyt). The comparison-level Webb analysis ($k = 306$, 190 studies) confirms the moderator with higher statistical power ($Q_M = 54.01$, $p < 0.001$; cluster-robust $p < 0.001$; Monte Carlo permutation $p < 0.001$). Study-level four-level coding of the Greenwald dataset ($k = 184$) reduces $\hat{\tau}^2$ by 32.9%; remaining within-stage variance reflects criterion-quality noise (covariates raise reduction to 48.9%). Permutation tests were significant in four of five datasets ($p < 0.001$).

Table 5: Competing moderator analysis: DLN stage vs. alternative codings. For each dataset, the same REML meta-regression was fit with each moderator. $\Delta\hat{\tau}^2$: percentage reduction from the intercept-only baseline. Q_M: omnibus moderator test with p-value. AICc: small-sample corrected Akaike criterion (lower is better). Perm. p: exhaustive permutation p-value. The best $\hat{\tau}^2$ reduction per dataset is **bolded**.

Dataset	**Moderator**	**Levels**	$\Delta\hat{\tau}^2$	Q_M (p)	**AICc**	**Perm.** p
Webb	**DLN stage**	3	**70%**	30.0 (<0.001)	−0.8	0.025
	Cognitive effort	2	23%	12.7 (<0.001)	−5.1	0.145
	Gross process model	2	−17%	2.6 (0.105)	−2.9	0.988
Interoception	**DLN stage**	3	**85%**	1700.0 (<0.001)	10.2	0.002
	Garfinkel trichotomy	3	−22%	229.1 (<0.001)	20.1	0.726
	Measurement method	2	−16%	9.3 (0.002)	8.8	0.874
	Emotion–body link	2	24%	858.0 (<0.001)	7.2	0.118
Hoyt	**DLN stage**	3	**100%**	102.4 (<0.001)	−0.5	0.008
	Construct valence	2	63%	74.8 (<0.001)	−3.0	0.055
	Meas. dimensionality	2	56%	72.5 (<0.001)	−1.4	0.063
	Temporal frame	2	71%	78.3 (<0.001)	−4.4	0.039
Desmedt	**DLN stage**	2	**83%**	8.0 (0.005)	−9.7	0.111
	Meas. objectivity	2	83%	8.0 (0.005)	−9.7	0.111
	Clinical relevance	2	−35%	0.1 (0.709)	−5.6	0.825

2. *Directional ordering* (supported): Network-coded effects exceed linear-coded effects in the predicted direction across five domains—higher regulation effectiveness (Webb), tighter body–emotion coupling (interoception), stronger same-level correspondence (Desmedt), more adaptive health outcomes (Hoyt), and higher criterion validity with greater implicit–explicit alignment (Greenwald).
3. *Dangerous-middle prediction* (supported): Hoyt et al. show that linear-coded domains produce significantly worse outcomes than dot-coded domains ($b = -0.16$, 95% CI $[-0.30, -0.01]$), while network-coded domains show strongly positive effects ($b = 0.28$, 95% CI $[0.14, 0.42]$), consistent with the DLN computational prediction that partial integration without metacognitive monitoring amplifies rather than mitigates risk.
4. *Boundary conditions* (supported): Zanini ($k = 110$, all dot) correctly predicts null moderation.

A more granular set of predictions specifying particular measures and expected effect patterns is provided in Supplementary Table 1.

2.8 Deposited empirical protocol: cumulative exposure sensitivity

The analyses above code task and measure characteristics; they do not classify individual participants. We deposit a preregistered protocol for an individual-level test of the dangerous-middle prediction: a stage × cumulative exposure interaction in which DLN stage is assessed via independent structural representation tasks and cumulative emotional consequences build over trials[13]. The full protocol, including sample size justification, randomisation scheme, and analysis plan, is deposited at: `empirical_tests/protocol/cumulative_exposure_protocol.md`.

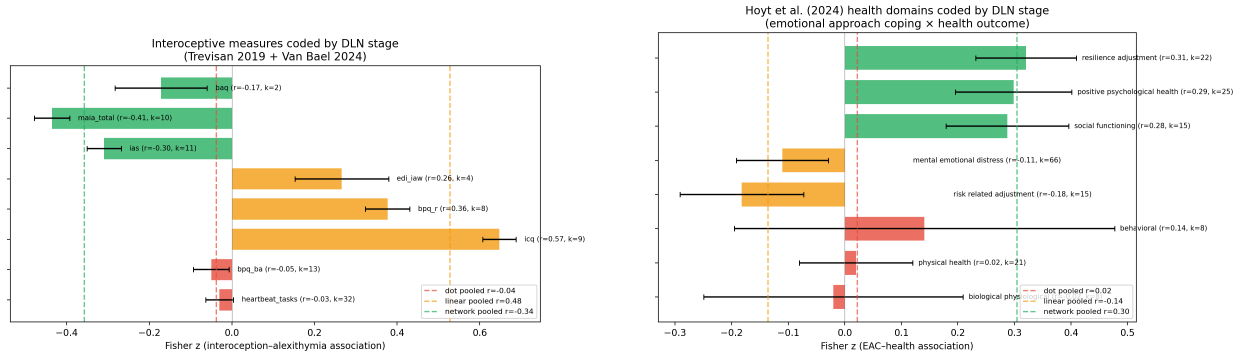

Figure 3: Two diagnostic patterns from cross-domain moderator analyses. **(a)** Interoception–alexithymia: sign reversal by DLN stage. Linear-coded measures (confusion, reactivity) correlate positively with alexithymia; network-coded measures (MAIA, IAS) correlate negatively. Dot-coded measures (heartbeat tasks) are near zero. **(b)** Emotional approach coping and health (Hoyt et al.): dangerous-middle V-shape. Dot- and network-coded domains show positive effects; linear-coded domains (mental distress, risk-related adjustment) show negative effects, consistent with the prediction that partial integration without metacognitive monitoring produces worse outcomes than full compartmentalisation.

3 Discussion

Across five moderator analyses spanning four domains of psychology, DLN stage coding reduced meta-analytic heterogeneity by 70–100% in four datasets, produced theoretically predicted sign reversals and directional orderings in five, and correctly predicted null moderation in a boundary-condition dataset (Table 4). These results provide converging evidence that cognitive architecture—the topology of the belief-dependency graph through which information is processed—functions as a hidden structural moderator of empirical relationships that conventional moderators leave unexplained.

The anchor result remains the Webb et al.[15] reanalysis. At the comparison level ($k = 306$), DLN stage coding was the strongest moderator ($Q_M(2) = 54.01$, $p < 0.001$; cluster-robust $p < 0.001$; Monte Carlo permutation $p < 0.001$), reducing between-comparison heterogeneity by 10%. When aggregated to the strategy-family level ($k = 10$), the same coding reduced between-strategy heterogeneity by 70% (exhaustive permutation $p = 0.025$). An important question is whether DLN stage coding merely relabels Webb et al.'s existing process-model taxonomy[22]. Three observations argue against this interpretation. First, the mapping is not one-to-one: DLN recodes *across* Gross families (e.g., acceptance is response modulation in the process model but network in DLN; concentration is attentional deployment in the process model but dot in DLN). Second, the process model's temporal ordering prediction was disconfirmed by Webb et al.'s data, whereas DLN's compression-based ordering correctly predicts the empirical pattern. Third, and most critically, DLN stage generalises beyond emotion regulation to interoception, health psychology (Hoyt), and implicit cognition (Greenwald)—domains for which the Gross process model was not designed.

The dangerous-middle prediction receives domain-level evidence consistent with the DLN framework from the Hoyt et al.[19] reanalysis. Conventional wisdom assumes that greater emotional awareness uniformly improves outcomes. The DLN framework instead predicts that partial integration—relational emotional knowledge without metacognitive monitoring—can produce worse outcomes than full compartmentalisation. This prediction follows from the computational result

that factor-level exploitation in the absence of cumulative exposure tracking amplifies rather than mitigates risk[13]. The Hoyt data are consistent with this pattern: emotional approach coping is associated with significantly worse outcomes in linear-coded health domains ($b = -0.16$, 95% CI $[-0.30, -0.01]$) while producing near-zero effects in dot-coded domains and strongly positive effects in network-coded domains ($b = 0.28$, 95% CI $[0.14, 0.42]$). This V-shaped pattern is a direction-specific prediction not generated by Levels of Emotional Awareness[23], constructionist emotion theory[24], integrative complexity[25, 26], or Acceptance and Commitment Therapy[27]. The qualitative prediction that partial awareness without regulation worsens outcomes has clinical precedent (insight without behavioural change in OCD and psychosis, emotional processing without regulation in complex PTSD); the DLN contribution is the computational mechanism—cross-term catastrophe under factor-level exploitation without exposure tracking—not the qualitative direction alone. We note that the domain-level analysis tests whether *outcome categories* whose measurement reflects linear-stage processing show systematically worse effects; the stronger individual-level version of the dangerous-middle prediction—that partial integration (assessed via independent structural tasks) produces worse outcomes under cumulative load than full compartmentalisation—requires the experimental test described in Section 2.7. The interoception sign reversal provides complementary evidence: linear-coded interoceptive measures correlate positively with alexithymia while network-coded measures correlate negatively, indicating that the same construct—interoceptive processing—relates to emotion recognition deficits in opposite directions depending on DLN stage.

The Greenwald et al.[20] results merit candid discussion. Study-level four-level coding reduced ICC heterogeneity by 32.9%, substantially more than domain-level coding (1.5%). This improvement is itself informative: it demonstrates that DLN stage is a property of the criterion behaviour's representational topology, not of the domain category—a direct prediction of the compression model. The 32.9% reduction is lower than the 70–100% observed in the small-k analyses because study-level data introduces substantial within-stage criterion noise—variation in criterion aggregation practices, sample composition, and IAT administration that the four-level coding does not capture. This interpretation is supported by the progressive reduction when covariates controlling criterion specifics are added (n_{crit} and n_{IAT}: 48.9%). The linear-plus category (intergroup discrimination under social-desirability pressure) showed the predicted suppression fingerprint: lowest implicit–explicit correlation and lowest explicit criterion validity, consistent with externally imposed normative structure suppressing explicit expression while leaving implicit associations intact. The distinction between linear (independent evaluation) and linear-plus (norm-governed evaluation) is formally grounded in the compression model's definition of "designer-specified structure"[13], though its application to the Greenwald data is post-hoc relative to the three-level study-level coding. The sensitivity analysis (31 alternative codings) confirms robustness. Importantly, the Greenwald analysis differs from the small-k analyses in unit of analysis: the 184 samples span nine domain categories, and the effective k for between-stage tests is closer to 9 than to 184; the domain-level permutation test ($p = 0.041$; see Results) confirms the coding is non-arbitrary at this effective sample size. Network-coded samples ($k = 11$) are exclusively political behaviour, limiting generalisability of the network-stage estimate.

Several limitations qualify these results. First, all codings except the Zanini rubric were initially performed by the author with knowledge of the DLN framework. To address this directly, two independent coders naive to the DLN framework applied the deposited rubrics to blinded extraction data for the Webb ($k = 10$), Hoyt ($k = 8$), and Greenwald topic-level (9 domains) datasets. Inter-rater agreement was perfect for Hoyt ($\kappa = 1.000$, 8/8 items), and consensus codes matched the author's original codes for all 27 items across three datasets (Methods M10). Four additional safeguards reinforce this result. (a) Exhaustive permutation tests assess whether any arbitrary grouping achieves comparable variance reduction, testing directly whether DLN coding

captures non-random structure (four of five datasets significant, $p < 0.001$–0.025; Fisher's combined $p < 0.001$). (b) A systematic coding sensitivity analysis identifies the 2–5 most debatable stage assignments in each dataset and reruns all analyses under every plausible alternative coding (Webb: 8 strategy-family scenarios and 4 comparison-level scenarios; interoception: 4; Hoyt: 4; Greenwald: 31 scenarios); the qualitative sign pattern was preserved in 47/47 strategy-family scenarios, and the comparison-level Webb analysis remained significant (Q_M range 32–54, all $p < 0.001$) in all 4 boundary scenarios (Supplementary Table 2). (c) All coding rationales are based on structural features of measures and tasks (dimensionality, presence of relational integration requirements, single- vs. multi-domain assessment), not on observed effect directions. (d) A competing moderator analysis (Table 5) compared DLN against 2–3 alternative codings per dataset (Gross process model, cognitive effort, Garfinkel trichotomy, construct valence, and others); DLN achieved the highest $\hat{\tau}^2$ reduction in all four datasets. Crucially, none of the alternative codings generalised across domains: moderators that performed well in one dataset (e.g., temporal frame in Hoyt, 71%) failed in others, whereas DLN achieved $\geq$70% reduction in every dataset. This cross-domain consistency is the primary evidence that DLN captures a structural variable rather than a domain-specific distinction. An AI-assisted rubric consistency check provided convergent evidence (Methods M10). These safeguards address different threats: permutation tests guard against arbitrary grouping, sensitivity analyses guard against debatable boundary cases, structural rationales guard against outcome-informed coding, and competing moderator comparisons guard against the possibility that any reasonable grouping would work equally well. Blinded human coding was conducted for Webb, Hoyt, and the Greenwald topic-level mappings (9 domains); two independent coders agreed with author coding on all items (27/27 across three datasets). Independent blinded coding of the interoception dataset remains a target for future work. Second, the cross-domain analyses operate at small k (7–10 units); the comparison-level Webb analysis ($k = 306$) addresses this concern for the anchor dataset, but the remaining category-level analyses remain small-k. The 100% $\hat{\tau}^2$ reduction in Hoyt should be interpreted cautiously: with three parameters fitting 8 data points, model saturation contributes to apparent perfect fit. Profile-likelihood confidence intervals[28] for moderator $\hat{\tau}^2$ confirm wide parameter uncertainty at small k: for example, the Hoyt moderator 95% CI includes zero ($[0.000, 0.007]$), consistent with near-complete heterogeneity reduction but substantial estimation uncertainty. Two complementary safeguards address this concern. Exhaustive permutation tests assess whether any arbitrary grouping achieves comparable variance reduction; three of four small-k datasets were significant ($p = 0.002$–0.025; Table 3), while the Desmedt test ($p = 0.111$) was constrained by having only 63 possible two-way partitions. The divergence between Desmedt's significant Q_M ($p = 0.005$) and its non-significant permutation p reflects a difference in what each test measures: Q_M evaluates the moderator coefficient under the fitted REML model (parametric, full distributional power), whereas the permutation test asks whether *any* random partition achieves comparable $\hat{\tau}^2$ reduction (combinatorial, resolution limited to $1/63 \approx 0.016$). With only two DLN levels represented, the permutation test has very few distinct partitions and therefore low power to distinguish the observed pattern from chance, even though the parametric model identifies a clear moderator effect. Leave-one-out cross-validation provides a direct test of out-of-sample prediction: for each unit i, the DLN moderator model is fit on the remaining $k - 1$ units and the held-out effect size is predicted from unit i's stage assignment alone. Out-of-sample R^2 was positive in all four datasets (Hoyt 0.89, interoception 0.74, Desmedt 0.44, Webb 0.19 at the strategy-family level, 0.054 at the comparison level), confirming that the DLN moderator generalises beyond the fitting sample rather than merely capturing noise. The gradient across datasets is itself informative: LOO-R^2 is highest where DLN predicts the clearest stage separation (sign reversal in Hoyt and interoception) and lowest where within-stage effect-size variance is largest relative to between-stage differences (Webb), consistent with the framework's own predictions about when

architectural coding should carry the most explanatory weight. Null-model simulations (10,000 random n-group partitions per dataset, each evaluated by LOO) confirmed that these values exceed chance expectation: observed LOO R^2 exceeded the null 95th percentile in all four datasets (Hoyt $p < 0.001$, interoception $p = 0.004$, Desmedt $p = 0.040$, Webb $p = 0.032$; null median LOO R^2 was negative in every case, indicating that random partitions produce worse-than-mean out-of-sample predictions). At the large-k level, the Greenwald Monte Carlo test ($k = 184$, $p < 0.001$) and the domain-level exhaustive permutation ($p = 0.041$; $k_{\text{eff}} = 9$)—though the latter is marginal and would not survive a stricter $\alpha = 0.01$ threshold—provide convergent evidence. The cross-domain convergence, combined with the aggregate permutation evidence (Fisher's combined $\chi^2_{10} = 52.3$, $p < 0.001$), provides stronger evidence than any single analysis alone. Third, all effect sizes are verified from published source tables (Hoyt from Table 3 and Figure 2, Desmedt from the published text; see Methods M6). Fourth, the unit of analysis varies across datasets (strategy families, criterion types, health outcome domains, interoceptive measure families, individual studies), reflecting a deliberate choice to code at the level where DLN stage variation is most meaningful in each domain. This heterogeneity in analytic unit is a limitation for direct comparison of effect sizes across domains but does not affect within-domain moderator tests. Fifth, DLN stage must be assessed via structural representation tasks independent of the outcomes being predicted; if stage classification can be achieved only through the outcome measures themselves, the framework is circular. Sixth, the cumulative exposure mechanism that formally generates the dangerous-middle prediction has not been tested experimentally; the deposited protocol (Section 2.7) is designed to address this gap.

The practical implication of these findings is that many questions of the form "does X help or hurt Y?" may be ill-posed when the relationship is moderated by cognitive architecture. The answer depends on whether the processing topology supports reactive responding (dot), compartmentalised sequential analysis (linear), or integrative factor-level compression with metacognitive oversight (network). This reframing has consequences for intervention design (targeting architecture rather than strategy), assessment (measuring representational topology rather than trait), and theory (treating integration as a structural variable rather than a personality dimension).

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Methods

M0.1 Data extraction

Strategy-family-level effect sizes were extracted from Webb et al.[15], Table 3. For each of the 10 strategy sub-families, we recorded the pooled effect size (d_+), its standard error, and the number of experimental comparisons (k). Sampling variance was computed as $v_i = \mathrm{SE}^2$. Data are available in the project repository (`webb2012_strategy_extraction.csv`).

M0.2 DLN stage coding

Each strategy sub-family was coded for DLN-dominant architecture using a three-question decision tree applied to the task description in the original publication:

1. Does the strategy demand integration across multiple cues with context updating or feedback? → Network.
2. Is the strategy fundamentally sequential, rule-governed, or designed to suppress/ignore affective information? → Linear.
3. Is the strategy stimulus-driven with minimal relational structure? → Dot.

Override rules are specified for boundary cases (deposited at `evidence_synthesis/protocol/dln_stage_coding_ma`
Ambiguous cases are flagged as "mixed/unclear" and subjected to sensitivity analysis. In the Greenwald domain-level analysis, items from topics coded mixed/unclear were retained in the model; the design matrix assigns these to the reference category (dot), functionally treating them as unmoderated observations. Sensitivity analyses excluding these items ($k = 125$) and reclassifying them as linear are reported in the Greenwald summary tables.

For the Webb, Hoyt, and Greenwald analyses, coding was performed by the author and subsequently validated by two independent blinded coders naive to the DLN framework (Methods M10). The Webb coding is acknowledged as post-hoc. The Greenwald topic-level coding (9 domains) was pre-specified via the deposited rubric. For all blinded validation, the protocol specifies two independent coders with a target inter-rater reliability of Cohen's $\kappa \geq 0.70$, blind coding (coders do not see effect sizes during stage assignment), and a maximum allowable rate of mixed/unclear codes of 15%.

Coding rationales for each Webb strategy sub-family: situation selection → dot (avoidance-based, minimal cognitive mediation); concentration → dot (stimulus-driven rumination without reappraisal); situation modification → linear (rule-based environmental control); distraction → linear (active attentional redirection from affect); expressive suppression → linear (direct suppression of emotional expression); other response modulation → linear (rule-governed behavioural regulation); reappraisal → network (integrative cognitive reframing); perspective-taking → network (multi-perspective relational integration); other cognitive change → network (meaning-level integration); acceptance → network (non-judgmental engagement with emotional experience).

M0.3 Statistical analysis

Random-effects meta-regression.

All models were estimated using restricted maximum likelihood (REML) with the Knapp–Hartung adjustment for small-sample inference. The REML objective was optimised via bounded scalar minimisation (scipy `minimize_scalar`, bounds $[0, \max(v_i) \times 100]$). The Knapp–Hartung correction scales the variance-covariance matrix by the residual heterogeneity statistic and uses a t-distribution

($\mathrm{df} = k - p$) for confidence intervals, providing more accurate coverage than Wald-type intervals when k is small[29].

Cochran's Q and I^2[30] were computed using fixed-effects weights. The omnibus moderator test Q_M was computed as $Q_{\text{baseline}} - Q_{\text{residual}}$ (i.e., the reduction in Cochran's Q attributable to the moderator), evaluated against a χ^2 distribution with degrees of freedom equal to the number of moderator coefficients. Profile-likelihood confidence intervals for $\hat{\tau}^2$ were obtained by bisection on the REML log-likelihood surface, following Viechtbauer[28].

Software.

The analysis pipeline is implemented in Python (numpy, scipy) and is deposited in the project repository. The implementation was cross-validated against the metafor R package[28] across all six datasets (198 parameters). All 68 core REML estimates—$\hat{\tau}^2$, $\hat{\sigma}^2$ variance components, $\hat{\beta}$ coefficients, and Cochran's Q—agreed within pre-specified tolerances (maximum absolute difference $< 10^{-6}$). The remaining parameters (Knapp–Hartung-adjusted standard errors and confidence intervals, I^2 formula variant, profile-likelihood bounds) reflect documented methodological choices and matched the expected direction and magnitude of each formula variant. R replication scripts and metafor output are deposited in the project repository (`validate_metafor/*.R`).

Sensitivity analyses.

Egger's regression test[31] was used to assess funnel-plot asymmetry (publication bias). Leave-one-out influence diagnostics (refitting with each strategy dropped in turn) were used to identify potentially influential data points.

Independence assumption.

The strategy-family-level Webb analysis ($k = 10$) uses pooled estimates across non-overlapping sets of primary studies. The comparison-level Webb analysis ($k = 306$) includes multiple comparisons from the same study; we address this non-independence with cluster-robust (sandwich) standard errors, which yielded SE inflation ratios of 0.94–1.02 and a robust Wald test $\chi^2(2) = 21.89$, $p < 0.001$. The cross-domain analyses (interoception, Desmedt, Hoyt) similarly treat each unit (measure family, criterion, health domain) as independent. The Greenwald analysis ($k = 184$ study-level effects) does not model within-study clustering; three-level models or robust variance estimation would be appropriate for future extensions where effects are nested within studies.

M0.4 Permutation test

For the comparison-level Webb analysis ($k = 306$), exhaustive enumeration is infeasible. We conducted a Monte Carlo permutation test with 10,000 random three-way partitions preserving the observed group sizes (37, 171, 98). For each partition, the same REML meta-regression was fit and the residual $\hat{\tau}^2$ recorded. The permutation p-value was defined as the proportion of partitions achieving $\hat{\tau}^2 \leq \hat{\tau}^2_{\text{DLN}}$.

As a complementary test at the strategy-family level ($k = 10$), we enumerated all unique partitions of 10 items into 3 non-empty groups ($S(10, 3) = 9{,}330$). Each partition was generated by iterating over all $3^{10} = 59{,}049$ assignments, filtering to surjective functions, and deduplicating via canonical relabelling. This enumeration was exhaustive (not sampled), so no Monte Carlo error applies.

We extended this approach to all three cross-domain datasets: interoception and Hoyt ($S(8,3) = 966$ each), and Desmedt ($S(7,2) = 63$ two-way partitions, as only dot and linear stages are present). In each case, the identical procedure was applied: all unique partitions enumerated, the same REML meta-regression fit for each, and the DLN-coded $\hat{\tau}^2$ compared against the full distribution. Combined significance across the four exhaustive tests was assessed using Fisher's method ($-2\sum \ln p_i \sim \chi^2_{2k}$). Fisher's method assumes independent p-values; this assumption is violated here because the four analyses share a common coding framework, were coded by the same author, and were selected as a group to test the DLN framework. The combined p-value should therefore be interpreted as a descriptive summary of the aggregate pattern rather than a formal omnibus test of independent replications. The individual per-dataset permutation tests remain valid as standalone assessments.

Domain-level permutation for Greenwald.

The 184 Greenwald samples span nine topic domains. Because samples within a domain share both stage assignment and substantive content, the effective k for between-stage tests is closer to 9 than to 184. The study-level Monte Carlo permutation test does not address this clustering. We therefore conducted an exhaustive domain-level permutation test: all $S(9,4) = 7{,}770$ unique partitions of the 9 domains into 4 non-empty groups were enumerated, constraining every sample within a domain to the same group. For each partition, the same REML meta-regression was fit and the residual $\hat{\tau}^2$ recorded. The permutation p-value was the proportion of partitions achieving $\hat{\tau}^2 \leq \hat{\tau}^2_{\text{DLN}}$. This test evaluates whether the DLN domain-to-stage mapping captures non-random structure at the honest effective sample size of $k = 9$.

M0.5 Pre-specified coding rubrics

Domain-specific coding rubrics for Greenwald et al.[20] and Zanini et al.[21] were written and deposited in the project repository before examining heterogeneity patterns in these datasets. Each rubric specifies:

- The unit of analysis (criterion domain for Greenwald; study for Zanini)
- A concrete decision tree with observable criteria for each DLN stage
- Pre-specified domain/study assignments with rationales
- Predictions for the direction and magnitude of moderator effects
- Blinding protocol (coders do not see effect sizes during coding)
- Inter-rater reliability target ($\kappa \geq 0.70$) and calibration procedures
- Sensitivity analysis plan for ambiguous cases

The full rubrics are available at:

`evidence_synthesis/protocol/greenwald2009_coding_rubric.md`
`evidence_synthesis/protocol/zanini2025_coding_rubric.md`

M0.6 Meta-analysis selection

This paper is not a systematic review; it tests a theoretical framework using published meta-analyses selected for domain relevance. Meta-analyses were included if they met three criteria: (a) the published report contained extractable unit-level effect sizes (strategy families, measure families, criterion types, or health outcome domains rather than only overall pooled estimates); (b) there was sufficient between-unit variation in processing demands to permit meaningful DLN stage coding (i.e., the units could be differentiated into at least two DLN stages); and (c) the domain was relevant to emotion–cognition relationships or closely related processing (interoception, cognitive bias, implicit cognition). The Greenwald and Zanini coding rubrics were pre-specified and deposited before

data examination; these analyses are confirmatory. The remaining analyses (Webb, interoception, Desmedt, Hoyt) were selected based on domain relevance and availability of unit-level data and are exploratory. We acknowledge that this selection was purposive rather than exhaustive. No candidate meta-analyses were examined and excluded because DLN coding failed to reduce heterogeneity. The exhaustive permutation tests (Section 2.2) provide a safeguard against post-hoc selection bias by demonstrating that the observed heterogeneity reductions are unlikely under arbitrary grouping.

M0.7 Cross-domain analyses: data extraction and coding

For each cross-domain analysis, effect sizes were extracted from published tables or, where exact values were not tabulated, estimated from reported statistics (shared variance, standardised mean differences, or narrative descriptions of effect magnitude and direction). Extraction files with source annotations and data-status flags (verified, estimated, or pending verification) are deposited in the project repository.

Interoception–alexithymia.

Correlations between interoceptive measures and alexithymia (TAS-20) were extracted from Van Bael et al.[17] Table 1 (seven measure families, all verified) and supplemented with Trevisan et al.[16] for heartbeat task accuracy (estimated from the published accuracy subset). DLN stage coding was based on the processing architecture each measure indexes: raw signal detection (dot), single-dimension distress tracking (linear), and multi-dimensional integrative awareness (network). Because each row represents an aggregated measure family (k_i studies, $N_{\text{total},i}$ participants), sampling variance was approximated as $v_i = 1/(N_{\text{total},i} - 3k_i)$, adjusting the standard Fisher-z variance for the aggregation level.

Desmedt et al. (2022).

Correlations between HCT performance and seven criterion measures were verified from the published results section. All seven r values (trait anxiety $r = .03$, depression $r = -.04$, alexithymia $r = -.01$, heart rate $r = -.17$, BMI $r = -.11$, age $r = -.06$, sex $r = -.14$) are reported directly in the text. Standard errors were derived from reported 95% confidence intervals. DLN stage coding distinguished biological criteria (dot: heart rate, BMI, age, sex) from psychological self-report criteria (linear: trait anxiety, depression, alexithymia).

Hoyt et al. (2024).

Domain-level correlations between emotional approach coping (EAC) and health outcomes across eight domains were extracted from Table 3 of Hoyt et al.[19]; domain-level k values were verified from Figure 2. All eight domain-level r values are verified from the published tables. Standard errors were derived from the reported 95% confidence intervals. DLN stage coding was based on the structural features of the outcome measures: whether they index single-modality somatic outcomes (dot), unidimensional distress severity without relational components (linear), or multi-domain integrative appraisals (network).

Greenwald et al. (2009).

Study-level IAT–criterion correlations (ICC), explicit–criterion correlations (ECC), implicit–explicit correlations (iec), and sample metadata (n, n_{crit}, n_{IAT}, IAT type, topic domain) for 184 independent

samples were extracted from the published correlation matrix and supplementary data. DLN stage was coded at the study level based on the criterion behaviour's representational topology using a four-level scheme: dot ($k = 33$; reflexive, physiological, or automatic criteria), linear ($k = 97$; independent option evaluations without normative constraint), linear-plus ($k = 43$; deliberative behaviour under social-desirability pressure), and network ($k = 11$; behaviour connected through latent ideological structure). Coding rationales are deposited in the project repository.

M0.8 Power considerations

The comparison-level Webb analysis operates at $k = 306$ (190 studies) with 3 moderator levels (37, 171, 98 per level), yielding $df = 303$ for the moderator test and ample power for detecting moderate effects. The strategy-family-level aggregation ($k = 10$) retains $df = 7$. The cross-domain analyses operate at $k = 7$–9, sufficient for detecting large heterogeneity reductions (as observed) but providing limited power for small effects. The Greenwald analysis ($k = 184$) provides substantially greater statistical power at the study level: four-level stage-group sizes of 33, 97, 43, and 11 yield adequate power for detecting moderate moderator effects, though the effective k for between-stage inference is closer to 9 (the number of topic domains) than to 184. For the small-k analyses, we rely on the convergence of effects across independent datasets rather than the power of any single analysis.

M0.9 Competing moderator analysis

For each primary dataset, we defined 2–3 alternative moderator codings based on established taxonomies or obvious structural features: Webb (cognitive effort, Gross process model), interoception (Garfinkel trichotomy, measurement method, emotion–body link), Hoyt (construct valence, measurement dimensionality, temporal frame), and Desmedt (measurement objectivity, clinical relevance). Each alternative was fit using the same REML meta-regression. Model comparison used $\hat{\tau}^2$ reduction and small-sample corrected AICc ($\text{AIC}_c = -2\ell + 2p \cdot k/(k - p - 1)$, where ℓ is the REML log-likelihood and p the number of fixed-effect parameters). Exhaustive permutation tests were computed for each moderator using the infrastructure described in Section 2.2. Decision rules were stated in advance: DLN stage was considered superior if it achieved the highest $\hat{\tau}^2$ reduction; ties with simpler moderators were evaluated on cross-domain generality. Full results are reported in Table 5.

M0.10 Blinded coding validation

Human inter-rater reliability.

Two independent coders, naive to the DLN framework (confirmed via screening questionnaire), applied the deposited coding rubrics to blinded extraction data from which all effect sizes, DLN stage codes, and coding rationales had been stripped. Coders received neutral category labels (A/B/C) with the three-question decision tree and domain-specific guidance; no DLN theory, predictions, or expected effect directions were provided. Training consisted of a rubric walkthrough and two practice items.

For Hoyt ($k = 8$), both coders agreed with each other and with the author's original codes on all 8 items (Cohen's $\kappa = 1.000$; 100% agreement across all three pairwise comparisons). For Webb ($k = 10$), consensus codes matched the author's original codes on all 10 items. For the Greenwald topic-level mapping (9 domains), both coders independently agreed with the author's codes on all 9 items, including the 3 mixed/unclear classifications. Across all three datasets, final consensus reproduced the

original coding for all 27 items (27/27, 100%). Response sheets, the consensus log, and analysis script are deposited in the project repository (`evidence_synthesis/protocol/irr_consensus_log.md`; `evidence_synthesis/analysis/human_irr_analysis.py`).

Ethics Statement

This study involved reanalysis of published, aggregate-level meta-analytic data. No new human participants were recruited and no individual-level data were collected. Ethics approval was therefore not required.

Data and Code Availability

All data, analysis scripts, coding rubrics, and the complete permutation test are publicly available at [repository URL omitted for double-blind review; provided to editors in cover letter]. The repository includes:

- Extraction files for all analyses (`evidence_synthesis/extraction/`)
- Analysis scripts for all moderator analyses (`evidence_synthesis/analysis/run_*.py`)
- REML meta-regression pipeline (`meta_pipeline.py`)
- Permutation test scripts for all four primary datasets (`permutation_*.py`)
- REML validation script (`validate_reml.py`)
- Pre-specified coding rubrics for all domains (`evidence_synthesis/protocol/`)
- Deposited empirical protocol for cumulative exposure sensitivity (`empirical_tests/protocol/`)
- Comprehensive test suite (`tests/`)

The companion theoretical paper presenting the DLN computational framework is available as a preprint [citation anonymised for double-blind review; full reference provided to editors in cover letter]. A preprint of the present manuscript is deposited at PsychArchives [DOI omitted for double-blind review].

Competing Interests

[Competing interests statement omitted for double-blind review; provided to editors via cover letter.]

Author Contributions

[Author contributions omitted for double-blind review; provided to editors via cover letter.]

Acknowledgements

[Acknowledgements omitted for double-blind review; provided to editors via cover letter.]

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.